

Rita Pecuch, Clinical Data Science & Engineering Strategic Leader

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LINKS

[Portfolio](#), [LinkedIn](#)

PROFILE

Data science and software engineering strategic leader with expertise in clinical trial systems, process automation, and technical innovation. Proven track record of translating complex business challenges into scalable software solutions within regulated environments that accelerate clinical development and improve operational efficiency. Passionate about applying open-source technologies, artificial intelligence, and modern engineering practices to drive measurable business impact.

AREAS OF EXPERTISE

Computerized Clinical Data Systems & Validation	Software Engineering & AI-Enabled Solutions
Regulatory Compliance (21 CFR Part 11, GxP)	CI/CD Pipeline Design & Automation
CDISC Standards	Technical Leadership & Stakeholder Management
Data Integrity & Audit Trail Management	Data Pipelines, Transformation & Common Data Models

EMPLOYMENT HISTORY

Mar 2024 — Present	Senior Data Configuration Engineer, Takeda	Cambridge, MA - Remote
Technical leader driving software development, process transformation, and technical strategy for clinical data solutions. Leads cross-functional initiatives to accelerate trial start-up, implement scalable software engineering practices, and deliver automation solutions that improve operational efficiency and quality. • Led design and development of AI-enabled test data entry tool reducing data entry time by over 90%. • Led design and development of audit-ready reusable R code library for clinical data enablement teams reducing development timelines across portfolio. • In partnership with key stakeholders, led development of business-controlled guidance for clinical data science dashboard development within regulated environments, closing critical compliance gap. • Spearheaded implementation of software engineering best practices within clinical programming—including DevOps pipelines, automated testing, reusable code libraries, and standardized development processes—to shorten development cycles, improve code quality, and reduce repetitive work. • Partners proactively with business and technical stakeholders to identify unmet needs, define solution strategies, and deliver high-impact software tools that improve operational performance. • Leads the design, development, and operationalization of scalable software and automation solutions that modernize clinical data management processes, accelerate clinical trial start-up, and reduce manual effort by up to 90% per database change through improved workflows and automation. • Leads strategic adoption of AI-enabled solutions for clinical trial system setup, overseeing end-to-end implementation and designing initiative-specific prompt engineering and validation strategies to improve efficiency, quality, and scalability. • Supervises a team of three functional service providers (FSPs), providing technical leadership, project oversight, mentoring, and end-to-end delivery management from requirements through production deployment. • Champions process improvements across the software development lifecycle that strengthen audit readiness, including development of a business-controlled documentation that closed a critical compliance gap.		

Feb 2023 — Mar 2024	Data Configuration Engineer, Takeda	
	Engineer driving software engineering, automation, and process improvement for clinical trial data systems. Partnered with engineering and business stakeholders to implement new technologies, establish scalable development practices, and deliver reusable data solutions that improved operational efficiency, documentation quality, and cross-trial consistency. • Served as the engineering representative for implementation of a new clinical data review platform, evaluating system capabilities and influencing standard operating procedures and upgrade impact assessments. • Partnered with cross-functional stakeholders to establish standardized, audit-ready documentation templates for clinical trial configuration, improving consistency and regulatory compliance. • Identified opportunities to automate manual trial configuration workflows using open-source technologies, reducing repetitive effort and leading implementation of a code version control process to support scalable team adoption. • Developed, peer reviewed, and maintained code for clinical trial data transformations and data visualizations, ensuring accuracy, quality, and adherence to appropriate clinical data standards. • Championed adoption of open-source tools by mentoring team members, developing technical guidance, and building team proficiency in new programming workflows.	
Jun 2019 — May 2022	Lead Medical Assistant, Aljundi Medical	Waterford, MI
	Directed a full-scale assessment and overhaul of medical office procedures, redesigning clinical and administrative workflows to eliminate inefficiencies, improve team productivity, and enhance patient care delivery. • Assessed and implemented business and technology solutions, optimizing the electronic health record (EHR) system to reduce staff workload, increase productivity, and maintain on-time operations during staffing shortages. • Revamped billing workflows and payment collection processes, increasing collections by approximately \$200 per week through improved accuracy, efficiency, and follow-up procedures. • Interviewed and trained new staff, establishing consistent workflows and best practices. • Translated complex scientific and medical information into clear, patient-friendly explanations to support informed decision-making. • Supported physicians by scribing during patient encounters, performing diagnostic testing, recording vital signs, and assisting with clinical examinations.	

EDUCATION

Aug 2024 — Present	M.S., Bioinformatics, Brandeis University	Waltham, MA
Sep 2022 — Nov 2022	Certificate, Software Engineering, Rutgers University	Camden, NJ
Sep 2017 — May 2021	B.S., Biomedical Sciences, Oakland University	Rochester, MI

PROFESSIONAL ORGANIZATIONS

Mar 2026 — Present	Project Lead, Ambassador Program, Healthcare Businesswomen's Association	
	As part of a leadership development program, leads planning and execution of enterprise-wide improvement initiative.	
Jan 2026 — Present	Organizing Committee Member, R/Pharma	
	Influences and helps selects topics included at industry events. Engages in hot topics and brings solutions back to team.	
Dec 2025 — Present	Treasurer, National Leiomyosarcoma Foundation	
	Develops automation tools to alleviate manual burden in bookkeeping process.	
Jul 2025 — Present	Git for Statistical Programming Working Group Member, PHUSE	

Discusses industry-specific challenges of Git implementation and contributes to consensus papers.

PRESENTATIONS

Jun 2026	Harnessing Generative AI to Improve Test Data Quality and Efficiency in Clinical Trial Set-Up Presented at R/Pharma GenAI Day	Virtual
Mar 2026	Harnessing AI: Enhancing Test Data Quality and Efficiency in Clinical Trial Set-Up Presented at PHUSE US Connect	Austin, TX
Mar 2026	The Use of Git in Statistical Programming Presented at PHUSE US Connect	Austin, TX
May 2025	Automating Clinical Test Data Generation and Entry with Generative AI and Innovative Technology Presented at PHUSE Computational Science Symposium	Utrecht, The Netherlands